

# Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

August 7th, 2026



2026-07-27

Thesis Presentation

Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

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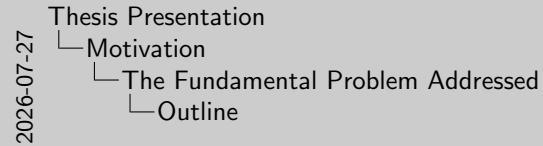
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1. Supervisor - Martin Lindström
2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

- 1 Motivation
  - The Fundamental Problem Addressed
  - Previous Work
- 2 Results and Contributions
  - Main Results
  - Key Ideas Behind Implementation
- 3 Summary
  - Takeaways
  - Further Reading



| Outline                   |                                     |
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| Motivation                | • The Fundamental Problem Addressed |
|                           | • Previous Work                     |
| Results and Contributions | • Main Results                      |
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|                           | • Further Reading                   |

# Stress Hyperglycaemia in ICU Patients

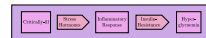
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## Thesis Presentation

### └ Motivation

### └ The Fundamental Problem Addressed

### └ Stress Hyperglycaemia in ICU Patients



1. Inflammatory response secretes stress hormones
2. Acute myocardial infarction, sepsis, stroke

## Addressing Hyperglycaemia

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      - The Fundamental Problem Addressed
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## Addressing Hyperglycaemia

- **Problem:**  
Stress-induced hyperglycaemia **increases** the likelihood of **mortality**.
- **A solution:**

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# Thesis Presentation

- └ Motivation

## — The Fundamental Problem Addressed

## Addressing Hyperglycaemia

- **Problem:**

Stress-induced hyperglycaemia **increases** the likelihood of **mortality**.

- **A solution:**

Maintaining glycaemic levels to 4-6 mmol/L reduced mortalities by 50% (Berghe et al. 2001).

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## Thesis Presentation

### └ Motivation

### └ The Fundamental Problem Addressed

### └ Addressing Hyperglycaemia

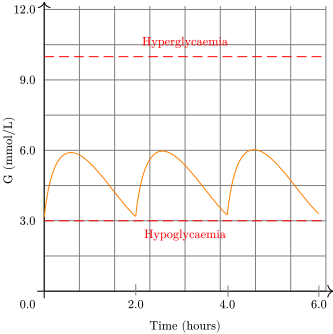
#### • Problem:

Stress-induced hyperglycaemia **increases** the likelihood of **mortality**.

#### • A solution:

Maintaining glycaemic levels to 4-6 mmol/L reduced mortalities by 50% (Berghe et al. 2001).

# Limitations of Intensive Insulin Therapy



- What:
- Why:

Figure: Tightly Controlled Blood Glucose

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Thesis Presentation

- └ Motivation
- └ The Fundamental Problem Addressed
- └ Limitations of Intensive Insulin Therapy

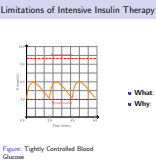


Figure: Tightly Controlled Blood Glucose

- What:
- Why:

# Limitations of Intensive Insulin Therapy

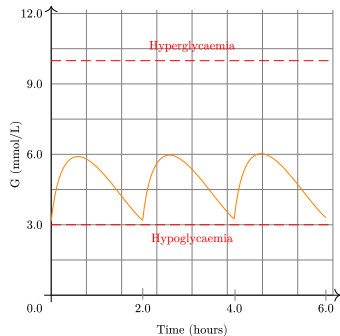


Figure: Tightly Controlled Blood Glucose

- **What:**  
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**

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└ Motivation

└ The Fundamental Problem Addressed

└ Limitations of Intensive Insulin Therapy



Figure: Tightly Controlled Blood Glucose

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# Limitations of Intensive Insulin Therapy

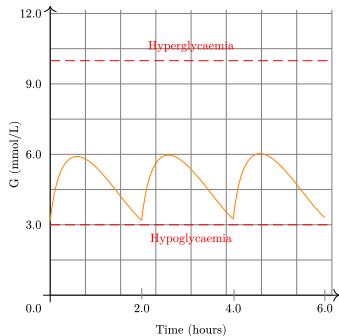


Figure: Tightly Controlled Blood Glucose

- **What:**  
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Solely utilizing insulin infusions struggles to address patient variability.

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## Thesis Presentation

### Motivation

### The Fundamental Problem Addressed

### Limitations of Intensive Insulin Therapy

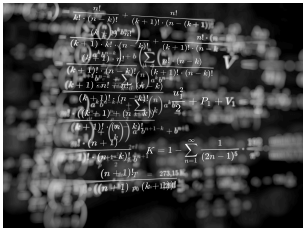


Figure: Tightly Controlled Blood Glucose

- **What:**  
Poor patient adaptability results in cases of **hypoglycaemia** and **high glycaemic variation**.
- **Why:**  
Solely utilizing insulin infusions struggles to address patient variability.

Infrequent measurements, not considering nutrition, and not complex to monitor dynamics changes in patient health

# Role of Model-based Glycaemic Control



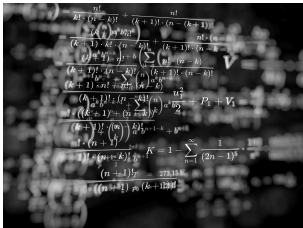
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    - The Fundamental Problem Addressed
      - Role of Model-based Glycaemic Control



Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

# Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.

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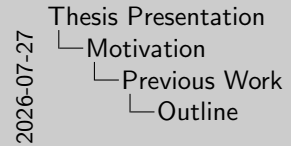
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  - └ The Fundamental Problem Addressed
    - └ Role of Model-based Glycaemic Control



Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

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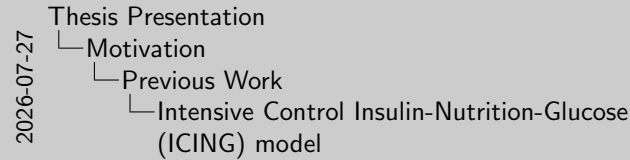
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# Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

| Dynamic                       | Notation    | Value    |
|-------------------------------|-------------|----------|
| Blood Glucose                 | $\dot{G}$   | mmol/L   |
| Interstitial Plasma           | $\dot{Q}$   | mU/L     |
| Plasma Insulin                | $\dot{i}$   | mU/L     |
| Glucose in the Stomach        | $\dot{P1}$  | mmol     |
| Glucose in the Gut            | $\dot{P2}$  | mmol     |
| Endogenous Insulin Production | $u_{en}(t)$ | mU/min   |
| Insulin Sensitivity           | $S_I(t)$    | -        |
| Input                         | Notation    | Value    |
| Exogenous Insulin             | $u_{ex}(t)$ | mU/min   |
| External Food                 | $D(t)$      | mmol/min |
| Parenteral Glucose            | $P_N(t)$    | mmol/min |



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- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic  $F$

$$\begin{aligned}t_k &= t_{k-1} + \Delta \\ \dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta} \\ F_k &\approx F_{k-1} + \Delta \dot{F}.\end{aligned}$$

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└ Motivation

└ Previous Work

└ Computational Approximation of the ICING model

1. According to Euler's method
2. Discretization error is bounded by  $\Delta$

- The ICING model equations are defined in continuous-time
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The discrete-time dynamic ICING model can be represented in the stochastic state-space.

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- Thesis Presentation
  - Motivation
    - Previous Work
      - Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

## State-space Representation

$$x_k \sim p(x_k \mid x_{k-1}, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

| System Variable | Notation |
|-----------------|----------|
| Dynamics        | $x_k$    |
| Inputs          | $u_k$    |
| Measurements    | $y_k$    |

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Bayesian Filtering

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1. The measurement often depends on the hospital, but will be discussed more later



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There are various methods of optimal filtering, and each has their own assumptions and limitations.

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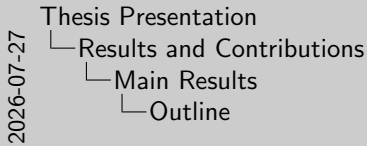
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1. The measurement often depends on the hospital, but will be discussed more later
2. Just the EKF and the PF will be in the topic of this thesis

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      - Extended Kalman Filter

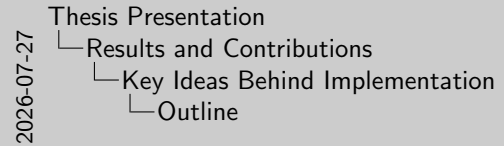
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 Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300.

 Lin, Jessica et al. (May 2011). “A physiological Intensive Control Insulin-Nutrition-Glucose (ICING) model validated in critically ill patients”. In: DOI: 10.1016/j.cmpb.2010.12.008.

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